Crisis Lens: An Overlooked Crisis Index for Humanitarian Funding Gaps

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ABSTRACT

Humanitarian funding remains misaligned with humanitarian need, and current public dashboards do not provide a compact, reproducible signal for prioritization. We present Crisis Lens, a system that defines and operationalizes an Overlooked Crisis Index (OCI) across crises, then connects that ranking to cluster-level gaps, project-level efficiency outliers, retrieval-based project recommendations, and a funding reallocation simulator. OCI combines people in need normalized by population, funding gap, and media neglect. Using UN HNO, FTS, population, and CBPF data from 2024 to 2026, we score 24 crisis contexts and analyze 8,091 projects. In 2026, mean year-to-date funding gap across 19 tracked crises is 0.917; because funding accrues throughout the year, this figure overstates full-year shortfalls. The five highest-OCI contexts are South Sudan (1.000), Yemen (0.858), Syria (0.844), Sudan (0.837), and Haiti (0.695). We identify 265 high-efficiency benchmark projects with a median beneficiary-to-budget ratio 15.6x above the remaining 7,826. The reallocation simulator models funding redistribution from better-funded crises to the most overlooked, turning OCI from a diagnostic into a prescriptive policy tool. A Databricks notebook reproduces the full pipeline for audit and reuse.

1 Introduction

Humanitarian systems routinely face the same operational failure: crises with high need and low coverage remain under-prioritized for extended periods (Van Wassenhove, 2006; Levine and Zervos, 2015). Decision-makers can access needs, funding, and project dashboards, but these signals are fragmented across sources and hard to compare in one view (Kovacs and Spens, 2007; Dubey et al., 2016). This fragmentation delays action where neglect is already compounding (Tarantola and Pupo, 2025; Tafese and Asfaw, 2025).

Crisis Lens addresses this gap with one measurable object and one deployable workflow. The measurable object is the **Overlooked Crisis Index (OCI)**, a country-year score that combines population-adjusted need, funding gap, and media neglect. The workflow ingests public humanitarian data, computes OCI, surfaces cluster-level funding bottlenecks, identifies project-level efficiency outliers, and returns similar high-performing projects through vector retrieval.

We frame the central pattern as the **overlooked crisis gap**: contexts with persistently high need exposure and weak funding coverage, often with low public attention (Levine and Zervos, 2015). This gap is visible in raw tables, but hard to rank and act on without a unified score and drill-down tooling.

This paper makes four contributions. First, we define OCI, a normalized score for comparing overlooked crises across contexts. Second, we build a reproducible data pipeline that integrates UN-aligned sources

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and produces crisis-level, cluster-level, and project-level analytics. Third, we deliver an interactive system for prioritization and action, including map-based ranking, crisis intelligence briefs, cluster diagnostics, efficiency outlier detection, vector-search recommendations, and funding-gap trend flags. Fourth, we provide a reallocation simulator that models the impact of redistributing funding from better-funded crises to the most overlooked, turning OCI from a diagnostic ranking into a prescriptive decision-support tool.

The remainder of this paper proceeds as follows. Section 2 formalizes the data model and OCI. Section 3 describes the system pipeline and application components. Section 4 details evaluation setup, metrics, and analysis protocol. Section 5 presents global, cluster, project, and trend results. Section 6 discusses related work and limitations. Section 7 concludes with deployment implications and next steps.

2 Measuring Overlooked Crises

2.1 Data Sources

We integrate public UN datasets spanning needs, funding, population, and project-level allocations (UN-OCHA HNO, 2026; UNOCHA FTS, 2026; UNOCHA COD-PS, 2026; UNOCHA CBPF, 2026). HNO provides people-in-need estimates, FTS provides requirements and funded amounts, COD PS provides population baselines, and CBPF provides project budgets and beneficiaries. We add a media-attention proxy from Google Trends when available (Table 1).

2.2 OCI Definition

OCI is computed per country-year and min-max normalized to $[0, 1]$ across tracked crises. Let P denote people in need, N population, G funding gap, and M media neglect. We compute

$$OCI = \left(\frac{P}{N}\right) \cdot G \cdot (1 + 0.2M)\,. \tag{1}$$

Funding gap is

$$G = 1 - \frac{\text{funding}}{\text{requirements}}, \tag{2}$$

clipped to $[0, 1]$. An earlier formulation included a severity proxy derived by quantile-binning the P/N ratio itself; we removed it because the derived severity correlates 0.96 with P/N, adding no independent signal and effectively squaring the need term. OCI scores are relative within the dataset: min-max normalization means that adding or removing country-year observations changes all scores.

2.3 Media Attention Proxy

Media attention is estimated from Google Trends queries of the form "[country] crisis" over the prior 12 months. A static baseline snapshot from January 2025 is substituted when live retrieval fails; the results reported in this paper use the static baseline. Normalized interest is inverted so that higher M signals lower attention and greater neglect pressure on OCI. The coefficient 0.2 in Equation 1 down-weights media relative to structural factors (need and funding); the ranking is stable across values from 0.1 to 0.3 (Appendix B).

Together, these three components produce a single country-year score that is interpretable, reproducible from public data, and decomposable into operational sub-signals at the cluster and project level.

Databricks notebook screenshot placeholder

Figure 1: Reproducibility pipeline. Databricks notebook used to reproduce pipeline execution and visual outputs.

3 System and Implementation

The Crisis Lens pipeline is implemented in Python and runs in Databricks or local mode. It ingests raw sources, harmonizes schema and country-year keys, computes OCI, and exports processed tables for the Streamlit frontend. The interface exposes six pages: global map, crisis drilldown, efficiency outliers, project recommender, funding forecast, and reallocation simulator.

Each crisis drilldown includes a **crisis intelligence brief**, a template-generated natural-language summary that synthesizes OCI components, severity classification, media visibility, and the most underfunded cluster into a 3-sentence situational overview. Severity tiers are derived from the P/N ratio: Extreme (> 0.30), Severe (> 0.20), Serious (> 0.10), Stressed (> 0.05), Minimal (otherwise). These tiers serve as display classifications and are not fed back into the OCI formula, avoiding the double-counting issue documented in Section 2.

The **reallocation simulator** models the impact of redistributing a user-specified percentage of funding from better-funded crises to the most overlooked ones. Funds are allocated proportional to OCI score and capped at each recipient's shortfall. A sensitivity sweep across reallocation levels (0–30%) shows the marginal gap reduction and estimated additional people reached. This turns OCI from a diagnostic score into a prescriptive decision-support input.

The recommender retrieves projects using cosine similarity over vectors composed of cluster type (one-hot), organization type (one-hot), log budget, and log beneficiaries. Actian VectorAI DB serves as the primary index, with a deterministic fallback to scikit-learn cosine similarity for reproducibility. This design decouples retrieval from a single vector backend, allowing reproduction with either engine and the same ranked output.

4 Experimental Setup

Evaluation covers three years (2024–2026), 24 crisis contexts, and 8,091 CBPF projects from 22 countries with active pooled funds. Table 1 summarizes dataset coverage. Four levels of output are analyzed: (1) country-year OCI ranking, (2) cluster-level funding gaps, (3) project-level efficiency outliers, and (4) funding-gap trend flags.

Three primary metrics are reported. **Funding gap** is unmet requirements as a fraction of total requirements. **Project efficiency** is beneficiaries per USD; outliers are identified via z-score on log-transformed efficiency ratios within each cluster-year partition, with a threshold of $z > 2.0$ for high-efficiency benchmarks. **Trend risk** is the slope of a linear fit over each country's full available funding-gap history, used as a descriptive risk flag.

Table 1: Dataset coverage used in Crisis Lens.

Dataset	Rows	Countries	Years
OCI scores (HNO + FTS + population)	65	24	2024 to 2026
FTS global funding	1,069	105	2000 to 2026
CBPF projects	8,091	22	2020 to 2026
Media attention scores	24	24	2025 baseline

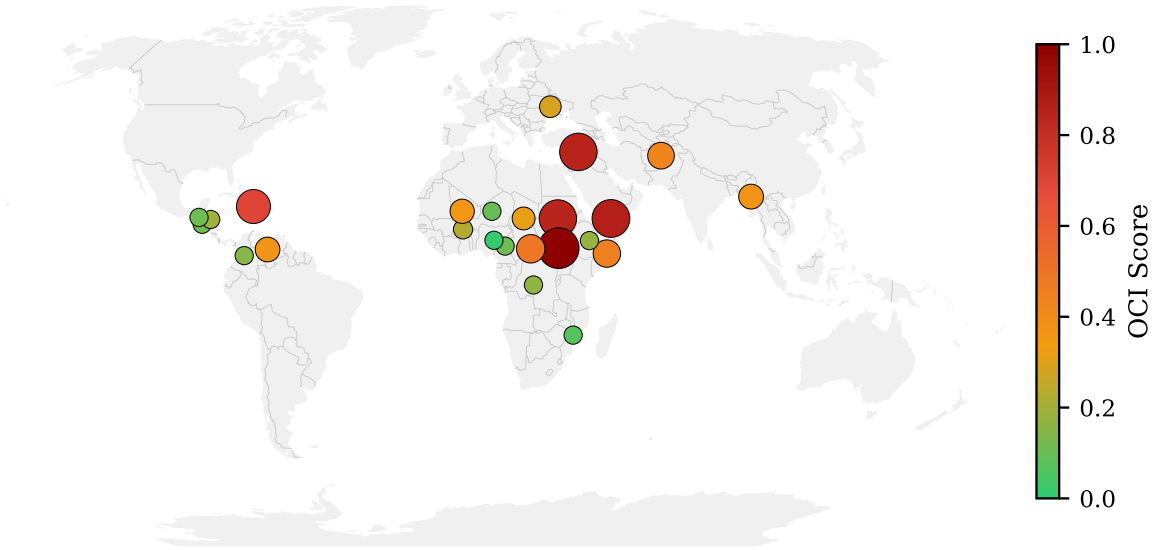

Figure 2: Global OCI map. Overlooked Crisis Index values for 2026. Higher values indicate greater neglect.

For recommender analysis, similarity search supports retrieval and qualitative transfer, not causal performance claims. For forecast analysis, extrapolations serve as early-warning indicators rather than predictive estimates. Because humanitarian plans set full-year requirements at the start of the year while funding accrues over subsequent months, 2026 funding-gap values reflect year-to-date disbursements at the time of data collection and are not directly comparable to final full-year figures from earlier years. These constraints keep interpretation aligned with available data volume and uncertainty.

5 Results

5.1 Global Ranking of Overlooked Crises

Figure 2 shows 2026 OCI values across tracked contexts. The ranking reveals need-coverage mismatches that single-source dashboards cannot surface. Table 2 reports the top five OCI scores. Across 19 tracked crises with 2026 data, mean year-to-date funding gap is 0.917 and median is 0.923. These figures reflect early-year disbursement status (Section 4) and will decrease as funding flows in over subsequent months.

5.2 Cluster-Level Funding Gaps

Cluster breakdowns localize where funding strain is most acute. For South Sudan in 2026, the largest gaps are Camp Coordination and Camp Management (100.0%), Logistics (99.3%), and Shelter and Non-Food

Table 2: Top OCI contexts for 2026. Funding gap is unmet requirements (%).

Country	OCI	Funding gap (%)	People in need (M)
South Sudan	1.000	89.5	9.9
Yemen	0.858	93.0	23.1
Syria	0.844	94.9	16.5
Sudan	0.837	87.0	33.7
Haiti	0.695	96.3	6.4

Items (98.1%). Figure 3 visualizes these bottlenecks; all three exceed 98%, indicating near-total absence of funded response in those cluster categories.

5.3 Efficiency Outliers and Benchmark Projects

At project level, we compute beneficiary-to-budget efficiency and detect high outliers within cluster-year partitions. We identify 265 benchmark projects from 8,091 total projects; these counts depend on keyword-based cluster assignment and would change with improved classification (Section 6). Benchmark median efficiency is 0.469 beneficiaries per dollar, while non-benchmark median efficiency is 0.030, a 15.6x difference. Figure 4 shows the distribution and outlier frontier.

5.4 Funding-Trajectory Risk Flags

We fit linear trends on each country's full funding-gap history (3–27 yearly observations depending on data availability) and project to 2027 as a directional risk signal. Among OCI-tracked crises with at least three years of data, the steepest statistically significant positive slopes belong to Myanmar (0.031/year, $n = 15$, $p < 0.01$), Cameroon (0.030/year, $n = 13$, $p < 0.01$), and Mali (0.025/year, $n = 20$, $p < 0.001$). South Sudan's full 16-year slope is 0.021/year ($p = 0.04$); the recent 2024–2026 spike is steep but not representative of the longer record. Figure 5 shows South Sudan's full funding-gap history. We treat these as directional risk flags, not forecasts of record.

5.5 Reallocation Simulator

The simulator models funding redistribution from crises with funding gap below a donor threshold to crises with OCI above a recipient threshold, allocated proportional to OCI score. At the default setting (10% reallocation from crises with < 50% gap to crises with OCI > 0.5), the model redistributes funding to the highest-need recipients, reducing average recipient funding gap by several percentage points. Figure 6 shows that the relationship between reallocation percentage and gap reduction is approximately linear up to 15%, after which marginal returns diminish as recipient shortfalls are increasingly capped.

This tool reframes OCI from a passive ranking into an active policy instrument. A fund manager can ask: "What happens if we redirect 5% from the best-funded plans?" and immediately see which crises benefit most and by how much.

6 Discussion

Humanitarian logistics research has consistently documented mismatch between assessed need and funded response (Van Wassenhove, 2006; Altay and Green, 2006; Kovacs and Spens, 2007; Dubey et al., 2016; Leiras et al., 2017). Recent funding-allocation studies further show that donor behavior tracks political salience and media visibility alongside need (Levine and Zervos, 2015; Tarantola and Pupo, 2025; Tafese

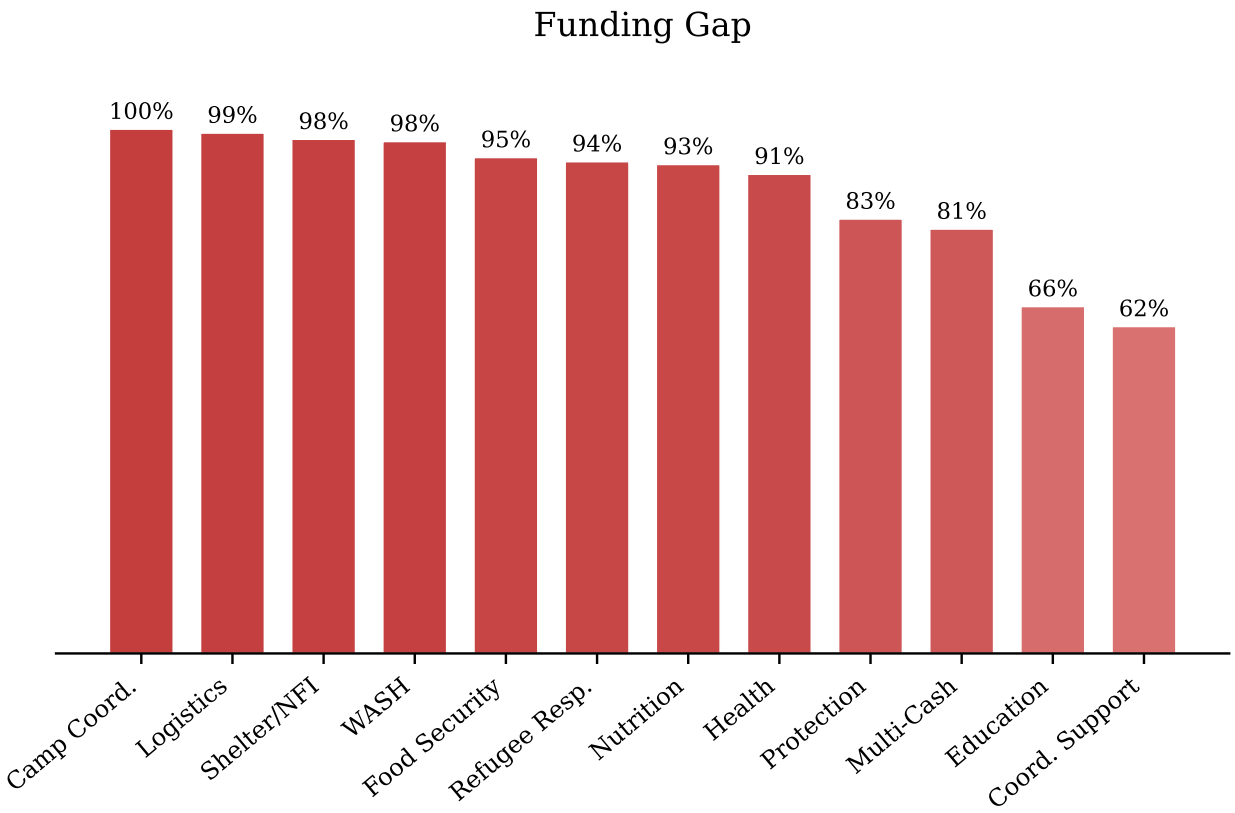

Figure 3: Cluster-level funding gaps for South Sudan (2026).

and Asfaw, 2025). OCI extends this line by turning neglect into a single comparative signal that can be decomposed operationally at cluster and project levels.

Crisis informatics has focused on event detection, social sensing, and multimodal classification for rapid situational awareness (Imran et al., 2015; Reuter and Kaufhold, 2020; Alam et al., 2018; Jabeen et al., 2024). Our target differs: we measure under-attended crises relative to need and funding, then support triage decisions. The media channel in OCI is therefore a visibility penalty, not a content-classification target.

Three limitations bound interpretation. Media attention from search interest is noisy and language-biased. At project level, cluster assignment for some CBPF entries relies on keyword inference and can misclassify ambiguous titles. Finally, linear trend extrapolation is descriptive, not predictive; slopes are sensitive to the time window used. These constraints limit causal claims and define concrete next steps.

The reallocation simulator adds a prescriptive dimension: rather than merely ranking neglect, it lets decision-makers model what-if scenarios for funding redistribution. The linear people-reached estimate is a deliberate simplification; real marginal returns vary by access constraints, absorptive capacity, and cluster-specific cost structures. We surface this as a modeling tool, not a policy directive, and include sensitivity analysis to show diminishing marginal returns at higher reallocation levels.

Crisis Lens identifies high-neglect contexts, localizes operational bottlenecks, surfaces transferable high-efficiency project patterns, and models reallocation impact—all from open humanitarian data. The outputs are usable for triage under current data constraints, and each limitation maps to a specific improvement path for future work.

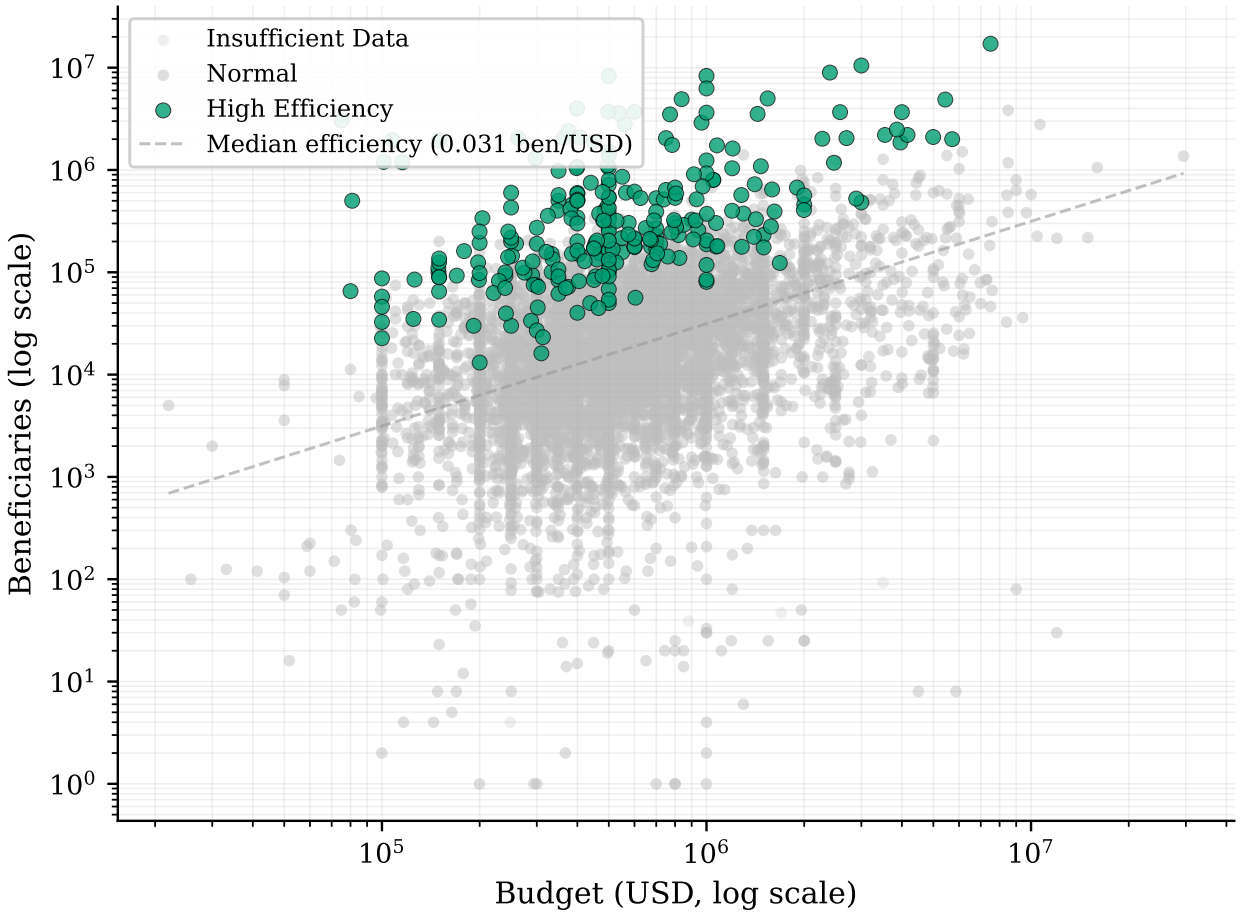

Figure 4: Project budget vs. beneficiaries. High-efficiency benchmark projects (green) cluster above the median efficiency line.

7 Conclusion

Crisis Lens provides a reproducible path from fragmented humanitarian data to ranked neglect signals and operational guidance. OCI identifies where population-adjusted need and funding shortfalls converge; cluster and project analytics then show where intervention design can change outcomes. The reallocation simulator extends this from diagnosis to prescription, allowing decision-makers to model the impact of funding redistribution before committing resources. Across 2026 results, underfunding is near-universal, and several high-OCI contexts are worsening under current trajectories.

Future work will expand temporal coverage for trend stability, incorporate richer need signals beyond population-adjusted estimates, add expert-grounded validation against practitioner prioritization, and integrate GDELT article-level media tracking to replace the Google Trends proxy. Crisis Lens gives decision-makers a ranked, auditable view of neglect built entirely from public data, designed to support faster and more defensible triage under humanitarian funding constraints.

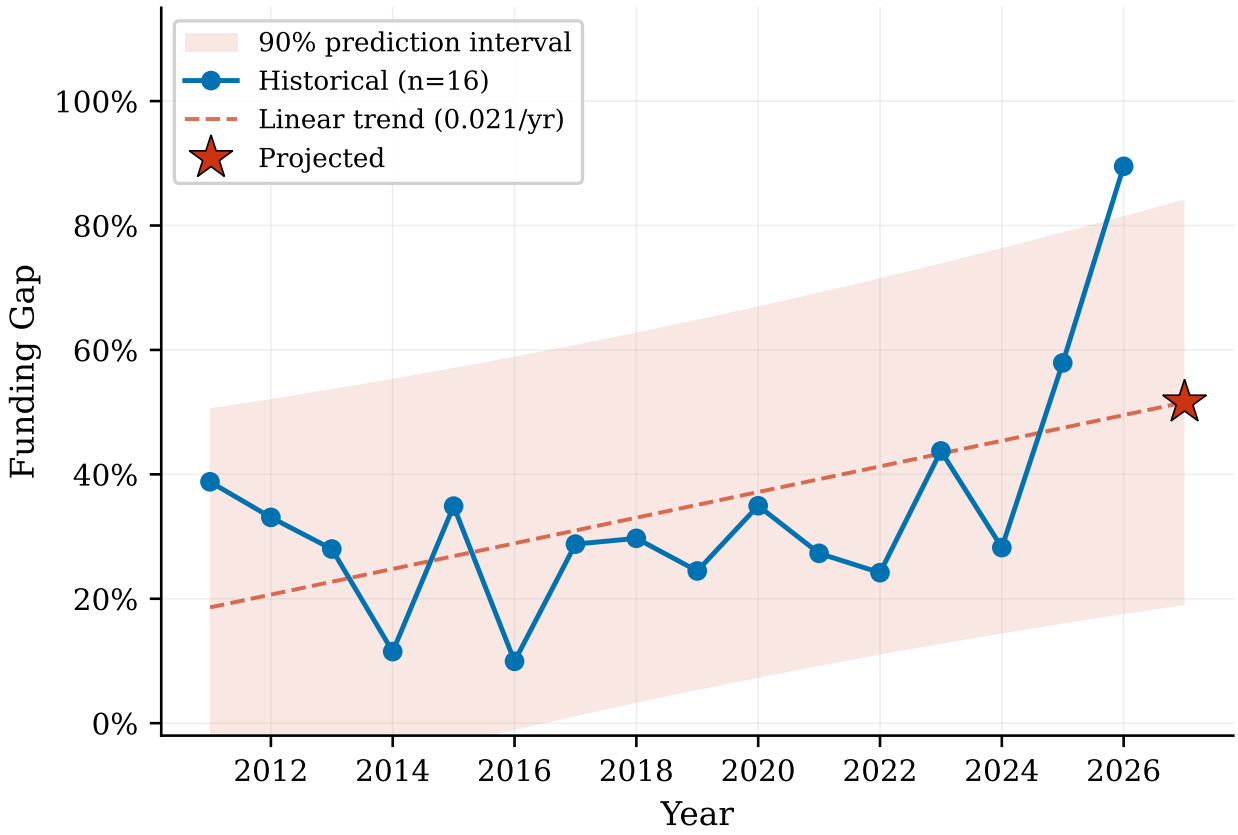

Figure 5: Funding-gap trend for South Sudan (2011–2026) with linear projection to 2027. Full 16-year history shows volatile but gradually rising gaps.

Reproducibility Statement

All data sources are public and listed in Table 1. The full pipeline is implemented in Python and runs locally or in Databricks. We provide a notebook that downloads source data, reproduces OCI computation, and regenerates paper figures. The Streamlit application consumes processed CSV outputs, and intermediate artifacts are stored in a versioned data directory.

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Reallocation sensitivity sweep placeholder

Figure 6: Sensitivity analysis: average recipient funding gap as a function of reallocation percentage. Diminishing returns emerge above 15%.

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Global Requirements and Funding (FTS). https://data.humdata.org/dataset/global-requirements-and-funding-data

COD PS global population data. https://data.humdata.org/dataset/cod-ps-global

CBPF data hub. https://cbpf.data.unocha.org/

A Complete 2026 OCI Scores

Table 3 reports all 19 country-level OCI scores for 2026, with decomposed components. P/N is people in need divided by total population, G is year-to-date funding gap, M is media neglect (1 = no search interest, 0 = high interest), and PIN is people in need in millions.

Table 3: Complete 2026 OCI scores with components. Sorted by OCI descending.

Country	OCI	P/N	G	M	PIN (M)
South Sudan	1.000	0.800	0.895	0.98	9.9
Yemen	0.858	0.671	0.930	0.90	23.1
Syria	0.844	0.710	0.949	0.37	16.5
Sudan	0.837	0.709	0.870	0.83	33.7
Haiti	0.695	0.539	0.963	0.78	6.4
Central African Republic	0.486	0.398	0.889	1.00	2.3
Somalia	0.452	0.389	0.867	0.86	4.8
Myanmar	0.370	0.300	0.930	0.88	16.2
Venezuela	0.366	0.290	0.963	0.80	7.9
Mali	0.354	0.285	0.919	0.99	5.1
Chad	0.314	0.250	0.934	1.00	4.5
Ukraine	0.282	0.293	0.865	0.00	10.8
Burkina Faso	0.223	0.187	0.912	0.98	4.4
DR Congo	0.164	0.145	0.897	0.95	14.9
Colombia	0.150	0.130	0.946	0.83	6.9
Cameroon	0.111	0.098	0.934	0.96	2.9
Niger	0.105	0.099	0.887	0.99	2.6
Mozambique	0.070	0.067	0.951	0.93	2.4
Nigeria	0.014	0.027	0.923	0.74	5.9

Ukraine illustrates how the media component operates: despite high P/N and G, its $M = 0.00$ (highest search interest) keeps OCI below smaller crises with higher neglect scores.

B OCI Sensitivity to Media Coefficient

The media multiplier α in $OCI = (P/N) \cdot G \cdot (1 + \alpha M)$ is set to 0.2 in the main analysis. Table 4 shows that OCI rank order is stable across $\alpha \in [0, 0.5]$. Spearman rank correlation between $\alpha = 0.2$ and all other values exceeds 0.99.

The only rank swaps occur between adjacent countries with similar OCI values (Yemen/Syria at ranks 2–3; Myanmar/Venezuela at ranks 8–9). Even removing the media component entirely ($\alpha = 0$) preserves 17 of 19 rank positions, confirming that structural factors (P/N and G) dominate the index.

Table 4: OCI 2026 rank by media coefficient α. Rank 1 = most overlooked.

Country	$\alpha = 0$	$\alpha = 0.1$	$\alpha = 0.2$	$\alpha = 0.3$	$\alpha = 0.5$
South Sudan	1	1	1	1	1
Yemen	3	3	2	2	2
Syria	2	2	3	4	4
Sudan	4	4	4	3	3
Haiti	5	5	5	5	5
CAR	6	6	6	6	6
Somalia	7	7	7	7	7
Myanmar	9	8	8	8	8
Venezuela	8	9	9	9	9
Mali	10	10	10	10	10
Spearman ρ vs. $\alpha = 0.2$		0.998	1.000	0.998	0.998

C Full-History Funding-Gap Trends

Table 5 reports linear trend slopes computed from each country's full available FTS funding-gap history. Asterisks mark slopes significant at $p < 0.05$. Slopes are much smaller than those from the 2024–2026 window alone, reflecting the volatility of annual funding cycles.

Ten countries show statistically significant positive slopes ($p < 0.05$), ranging from 0.010 to 0.031 per year. Ukraine's slope (0.006, $p = 0.75$) is not significant over its 13-year history; the recent 2024–2026 increase reflects normalization from the exceptional post-invasion funding surge of 2022, not a novel deterioration.

D Recommender Example

The vector similarity recommender retrieves benchmark projects comparable to a given query. Features are: cluster type (one-hot), organization type (one-hot), log budget, and log beneficiaries, combined with cosine similarity.

As an example, querying a South Sudan Health project (budget $250K, 12,520 beneficiaries, efficiency 0.050 ben/USD) returns five Health benchmark projects from other contexts with 6–13× higher efficiency at comparable budget scales. All five recommendations have cosine similarity above 0.99, indicating close structural matches in cluster and scale. This demonstrates the recommender's ability to surface transferable project designs for underperforming contexts.

Table 5: Funding-gap trend slopes from full FTS history. n = years of data.

Country	Slope (/yr)	n	p
Myanmar	0.031	15	0.009*
Venezuela	0.030	8	0.233
Cameroon	0.030	13	0.004*
Mali	0.026	20	<0.001*
Yemen	0.025	19	0.002*
Chad	0.024	23	<0.001*
South Sudan	0.021	16	0.036*
Niger	0.016	22	0.008*
Haiti	0.015	20	0.016*
Sudan	0.012	27	<0.001*
Colombia	0.012	9	0.288
Burkina Faso	0.010	20	0.080
DR Congo	0.010	27	0.011*
Mozambique	0.010	12	0.419
Ethiopia	0.008	13	0.269
Afghanistan	0.007	23	0.298
Syria	0.006	18	0.413
Ukraine	0.006	13	0.752
El Salvador	0.003	8	0.590
Guatemala	0.003	9	0.831
Nigeria	0.003	14	0.856
Somalia	0.002	27	0.640
CAR	−0.002	24	0.741
Honduras	−0.005	8	0.803